Xucheng (Victor) CHEN

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EDUCATION

Chinese University of Hong Kong, Shenzhen **Shenzhen, China**
Economic Science in School of Management and Economics (SME) Sep 2023 – Present

- Major GPA: 3.970/4.00
- **Selected Coursework: (Achieved A in all SME courses, with multiple top-tier rankings)**
  - Intermediate & Advanced (PhD-level) Microeconomics/ Advanced Macroeconomics (Full marks in midterm exam)/ Game Theory/ Intermediate & Advanced (PhD-level) Econometrics
  - Quantitative Methods for Policy Evaluation/ Advanced Methods for Empirical Research (PhD-level) /Programming (Python)/ Machine Learning (4/258 in total grades)/
  - China Economics/ Advanced (PhD-level) Political Economics/ Industrial Organization (PhD-level)
  - Calculus (Top 10 in Calculus Ⅱ –Multiple Calculus)/ Linear Algebra (Full marks in final exam)/ Optimization (13/358 in midterm exam and 96/100 in final exam)/ Ordinary Differential Equations/ Real Analysis
  - Probability Theory/ Theory of Statistics/ Time Series
  - Gender Study/Study of Logics/ Intro to Sociology
- **Honors & Awards:**
  - National Scholarship (Highest Honor for the Undergraduates in China, 2025)
  - University Honors (Excellent Student Award, 2025)
  - 31st Undergraduate Research Award (URA), 2026
  - 30th Undergraduate Research Award (URA), 2026
  - 29th Undergraduate Research Award (URA), 2025
  - Curiosity Research Award, 2025
  - SME Academic Performance Scholarship (Top 2% in SME, All academic years, 2023-Present)
  - Dean’s List (All academic years, 2023-Present)
  - Ling Inspirational Scholarship (2023-2026), Bowen Ⅱ Scholarship (2023-2026)
  - Ling College Leadership Scholarships Award (2026)
  - Ling College Volunteer Scholarships Award (2025)
  - Annual Hot Search Award (SME News, 2025)
  - Person of the Year (interviewed by Lakeside Magazine, 2025)
  - Semifinalist and 4th Best Speaker (Open Group) in Greater Bay Area Novice Championship (GBANC), 2025
  - Second Prize in Guangdong Contemporary Undergraduate Mathematical Contest in Modeling (2024)

University of California, Berkeley **California, United States**
Berkeley Global Access Visiting Student Programs, Nominated by Home Institution Jan 2026 – May 2026

- **Selected Coursework:** Real Analysis (A); Advanced Econometrics Ⅱ (PhD-level, A-); Advanced Political Science Ⅱ (A, PhD-level)

University of Cambridge, Pembroke College **Cambridge, United Kingdom (Remote)**
Online Summer Research Programme (2025), Independent Research Supervised by Prof. Myungun Kim Jul 2025

- **Independent Paper:** Beyond Patents: R&D, Capital, and the Productivity Puzzle in Early-Stage High-Tech Firms (Awarded as Best Paper in this programme).

The London School of Economics and Political Science **London, United Kingdom**
Summer School Programme Jul 2024 – Aug 2024

- **Selected Coursework:** Introductory Macroeconomics (A, Rank First in this course)

RESEARCH ASSISTANT EXPERIENCE

Machine Learning for Chinese Court-Video Transcription and Role Attribution
Research Assistant (Supervised by Prof. Shaoda Wang, University of Chicago)
Berkeley, The United States (Remote) Jun 2026 – Present

- Audited and extended a two-stage machine-learning pipeline that converts Chinese court-hearing video and audio into timestamped, speaker-diarized transcripts and courtroom-role labels using ffmpeg, FunASR Paraformer, CAM++/3D-Speaker, and LLM-assisted classification.
- Stabilized batch orchestration and model interfaces; added structured logging, cache controls, and restart-oriented processing, and diagnosed timestamp, runtime, speaker-mixing, and empty-output failure modes rather than treating file existence as validated accuracy.
- Designed a 20–30-case pilot and an approximately 200-hearing gold-set evaluation plan covering CER, timestamp alignment, diarization error rate, role-attribution F1, runtime, failures, and cost; prepared a Windows/GPU handoff for scalable execution.

Speech Recognition and Treatment-Compliance Analysis in Congo
Research Assistant (Supervised by Prof. Clotaire Weigel, UC, Berkeley)
Berkeley, The United States May 2026 – Present

- Developed an end-to-end workflow for low-resource field-sermon audio, including long-recording chunking, automatic speech recognition, language detection, and preparation for treatment-compliance analysis.
- Tested Whisper and Meta MMS for French and Tshiluba-related audio; evaluated a Luganda proxy for Tshiluba when direct model support was unavailable and documented model uncertainty and limitations.
- Translated administrative, collective, and individual treatment messages into structured coding dimensions covering tax communication, justice and poverty, prosperity, and individual responsibility for downstream text analysis.

Technology-Driven Market Concentration through Idea Allocation
Research Assistant (Supervised by Prof. Shaoshuang Yang, CUHK-Shenzhen and Prof. Yueyuan Ma, UC, Santa Barbara)
Shenzhen, China Jul 2025 – Sep 2025

- Helped in constructing Herfindahl–Hirschman Index (HHI) measures of market concentration and translated established algorithms from the literature into reproducible Stata code.
- Processed U.S. patent data from the USPTO to systematically calculate backward- and forward-citation indices; reviewed patent documents, citation records, and related literature to characterize technology novelty and idea quality.
- Built a patent citation graph linking citing and cited patents and implemented graph-based measures of centralization and decentralization; proposed algorithmic refinements and documented the empirical workflow.

Collaborative Growth and Global Win-Win: An Overview of China's Key Industry Clusters Going Global
Research Assistant (Supervised by Prof. Jian Wang and Prof. Jieshuang He, CUHK-Shenzhen)
Report Author (Center for China's Economic and Outbound Strategy, Shenzhen Finance Institute)
Shenzhen, China Jun 2025 – Aug 2025

- Co-authored a comprehensive research report focusing on outbound expansion strategies of China’s pan-cultural industry cluster. Synthesized insights from 300+ reports and case studies to identify success patterns, structural challenges, and strategic policy recommendations.
- Analyzed global expansion paths of Chinese vs. foreign firms, including full value chain dynamics across upstream, midstream, and downstream sectors. Produced policy suggestions and economic visualizations; integrated cross-industry research into a unified strategic narrative.
- Engaged in collaborative discussions with the Global Relations Project Team from Southern California to contextualize findings within broader international frameworks.

The Obstacles Faced by Female Entrepreneurs When Attempting to Recruit Talent
Research Assistant (Supervised by Prof. Zhongchen Hu, CUHK-Shenzhen)
Shenzhen, China Aug 2024 – Mar 2025

- Conducted a literature review to show the backgrounds of obstacles in the specific jobs, analyzing the phenomenon of internalization of discrimination in the recruit process.
- Conducted and operated field experiments online via the LinkedIn platform; maintained personal platform profiles to ensure the validity and integrity of the field experiment.
- Collected responses and data from target employers to companies that are identical in all basic characteristics except for the founder's gender, in order to quantify the employment situation of women in the corresponding industry.

RESEARCH EXPERIENCE

Robust Trust with Multiple Advisers
Working Paper, Independent Research (Supervised by Prof. Bin Liu, CUHK-Shenzhen)
Shenzhen, China Aug 2026 – Present
Accepted by GBA AI Alliance Conference (GAAC 2026)

- Developed a robust decision model with K potentially misaligned, strategically coordinated advisers and proved that positive worst-case value is possible if and only if each adviser is aligned with probability above one half for any finite K.
- Under symmetric posteriors and quadratic loss, derived the exact margin-ladder aggregation rule and arbitrary-K saddle point; built a deterministic 24-case audit with independent linear-programming cross-checks for $K \leq 3$ and explicit theorem boundaries.

Who Gets Cut Out? Generative AI and the Reorganization of Knowledge Work inside Firms
Work in Progress, Undergraduate Author
Shenzhen, China Aug 2026 – Present

- Developing a China-first framework for how generative AI may change task bundles, handoffs, decision rights, and management layers; designed an observable-responsibility taxonomy for front/product-facing, middle/intermediary, and back-office work while separating exposure, adoption, task change, and labor outcomes.
- Building a zero-new-cost empirical plan around verified adoption events, public listed-firm outcomes, and full-text vacancies only where licensed access is confirmed, with validation, pre-trend, placebo, and alternative-taxonomy tests specified before causal claims.

Who Moves Up? Entrepreneurial Risk Preferences and Small-Firm Trajectories
Working Paper, Undergraduate Author (With Prof. Shaoshuang Yang, CUHK-Shenzhen)
Shenzhen, China May 2025 – Present
Sponsored by 31st Undergraduate Research Award (URA)

- Co-developed an empirical study linking 4.37 million Alipay merchant-months for 100,000 micro-merchants (2020–2023) to financial-suitability risk ratings; built mobility, tail, volatility, credit, portfolio, and lockdown-exposure measures and the 83,854-merchant 2020-baseline trajectory analysis.
- Documented a 1.24-percentile higher 2023 sales rank and 1.87-percentage-point higher top-quartile entry probability per rating unit, conditional on baseline rank; estimated fixed-effects lockdown models and a leave-one-out IV sensitivity analysis while treating nonrandom-rating, pre-trend, and exclusion issues explicitly.

Mobile Internet and Crime: Evidence from China's 4G Rollout
Working Paper, Undergraduate Author (with Dr. Chenxuan Chen, Peking University)
Shenzhen, China Sep 2026 – Present

- Exploit the staggered rollout of 4G base stations across Chinese cities and counties in a difference-in-differences framework to identify the causal effect of mobile internet access on local crime rates.
- Construct a panel dataset linking 4G infrastructure deployment records with criminal conviction data from PKU LawSec ( ), a comprehensive repository of Chinese court judgments, to build a unique crime–connectivity panel.
- Find that expanded 4G coverage significantly increases local crime rates; document heterogeneous effects across crime categories including property crimes, violent offenses, and cyber-related criminal activity.
- Investigate mechanisms through which mobile internet access affects criminal behavior, including information diffusion, reduced coordination costs among criminal networks, and reallocation of economic opportunity.

Factory Parents, Classroom Outcomes: Occupational Networks and Peer Effects in Education
Working Paper, Independent Research (Advisor: Prof. Emily Zheng, CUHK-Shenzhen)
Shenzhen, China Dec 2025 – Present
<u>*Accepted by the 2nd Chinese Educational Year Seminar, Sponsored by 29th & 30th URA, Curiosity Research Award*</u>

- We study how exposure to classmates from manufacturing and production worker families affects academic achievement in urban Chinese middle schools. Exploiting the random assignment of students to classrooms within school-grade cohorts in the China Education Panel Survey (CEPS), we find that a ten-percentage-point increase in the share of peers with production worker parents reduces standardized academic scores by approximately 0.21 standard deviations.
- The effect is robust and statistically significant across all specifications, and present for all three core subjects (mathematics, Chinese language, and English). We investigate mechanisms using rich parent questionnaire data. Exposure to the working-class peer families significantly reduces other parents' engagement with children's friendship networks, particularly the inter-family social connections and peer-relationship communication that require bridging social capital, as well as school discussion frequency, parental confidence in children's academic futures, and parent-initiated teacher contact.
- Parents in classrooms with more working-class families also report systematically lower evaluations of the educational environment. Falsification tests using business and service sector workers, who share comparable earnings and education levels but work in more socially interactive occupations, yield uniformly null effects across all outcomes. These findings reveal how occupational class composition of the classroom shapes the parental social capital networks that support children's education, with implications for school integration policies and class-based educational inequality in rapidly industrializing economies.

Retail Money Flows and Stock Returns in China
Working Paper, Undergraduate Author (with Prof. Chuan Shi, CUHK-Shenzhen)
Shenzhen, China May 2025 – Present

- Contributed to a quantitative asset pricing paper investigating the predictive power of small-order net inflows on stock returns using Chinese high-frequency data. Constructed small-order flow indicators from tick-by-tick transaction records (2018–2024), including rigorous definition of trade directionality and size thresholds (λ-parameter calibration), with reference to but beyond the granularity of industry reports (e.g., Kaiyuan Securities).
- Conducted Fama-MacBeth regressions across multiple λ and net flow estimation methods (s2/s3), showing strong robustness of results; implemented portfolio sorts (univariate & double sorts) to test the return predictability of representative strategies (e.g., s2/λ3).
- Performed orthogonalization procedures to control for confounding effects and assess interactions between small-order signals and salient behavioral pricing factors such as the salience factor (ST); demonstrated significant alpha in double-sorted portfolios.
- Independently wrote core sections of paper including methodology, data construction, and empirical results; developed all analytical codes in Python and R, covering factor modeling, signal engineering, and high-frequency data cleaning.

When Structure Beats Flexibility: A Comparative Study of Forecasting in Bitcoin Markets
Undergraduate Independent Research (Advisor: Prof. Haichun Ye, CUHK-Shenzhen)
Shenzhen, China Sep 2025 – Dec 2025

- Conducted a unified empirical comparison of econometric volatility models (AR(1)–GARCH / TGARCH) and LSTM neural networks in forecasting Bitcoin returns and volatility using daily data (2015–2025) in EViews and Python.
- Established key stylized facts of cryptocurrency markets—nonstationary prices, stationary log returns, volatility clustering, heavy tails, and asymmetric shock responses—and implemented a full ARMA–(T)GARCH model selection and diagnostic pipeline (ADF tests, ARCH–LM tests, BIC-based selection, post-estimation Q-statistics).
- Designed a transparent 500-day out-of-sample forecasting framework with identical information sets across models; demonstrated that the predictive edge of TGARCH over LSTM arises from explicitly structured conditional variance dynamics rather than nonlinear mean modeling, with substantial gains concentrated in volatility forecasting.

A Bayesian Game of Online Accusations
ECON3160 Game Theory Course Project (Advisor: Prof. Yangbo Song, CUHK-Shenzhen)
Shenzhen, China Sep 2025 – Dec 2025

- Developed a Bayesian game of online accusations in which a Key Opinion Leader (KOL) with noisy private information strategically decides whether to accuse a target, while a continuum of users—observing only the public accusation—choose whether to join a collective attack or exercise restraint.
- Characterized Perfect Bayesian Equilibria under pooling and separating accusation strategies, identifying conditions under which platform amplification and weak reputational penalties generate self-reinforcing “digital witch-hunt” equilibria despite high uncertainty about guilt.
- Extended the baseline framework to richer informational and strategic environments, including (i) competing KOLs with independent signals and pile-on complementarities, and (ii) dynamic reputational penalties that accumulate over time, showing how corroboration, attention competition, and forward-looking discipline reshape equilibrium accusation standards and crowd behavior.

Beyond Patents: R&D, Capital, and the Productivity Puzzle in Early-Stage High-Tech Firms
Undergraduate Independent Research (Advisor: Prof. Myungun Kim, Cambridge)
Cambridge, United Kingdom (Remote) Jun 2025 – Aug 2025
Awarded as Cambridge Online Summer Research Programme (2025) Best Paper

- Conducted an independent empirical study on the link between innovation and productivity in Chinese high-tech startups, using a proprietary dataset (2020–2024) of patent records, R&D expenditures, and firm performance indicators.
- Employed quantitative analysis and mechanism-based reasoning to examine the signaling role of patents and the performance impact of R&D investment. Identified heterogeneity across industries and regions, highlighting the nuanced role of innovation in emerging markets.
- Awarded Outstanding Research Paper Prize; presented findings at the programme’s academic research conference. The paper can be viewed on website https://doi.org/10.48550/arXiv.2507.18227.

Sexual Minorities’ Perspectives on Feminism: A Dialogue Between Gender and Sexuality
Undergraduate Independent Research (Advisor: Prof. Shirley Xiao, CUHK-Shenzhen)
Shenzhen, China Sep 2024 – Mar 2025
Accepted by CUHK-Shenzhen Undergraduate Research Conference (2024)

- Investigated how overlapping identities shape group alignment and social preferences through semi-structured interviews and inductive thematic coding.
- Analyzed structural and normative constraints on identity-based coalition-building using intersectionality theory and comparative case analysis.
- Findings offer insights into preference formation, collective action dynamics, and the design of more inclusive social policies.

Research on Production Process Decision-making Based on Defect Rate Optimization
Undergraduate Author
Shenzhen, China Sep 2024
Second-prize paper in Guangdong Contemporary Undergraduate Mathematical Contest in Modeling

- Developed a sampling inspection model using one-sided hypothesis testing and statistical distributions, minimizing inspections while ensuring quality.
- Built a cost-revenue optimization model of the full production chain using Python, incorporating enumeration and genetic algorithms for optimal decision-making.
- Applied Monte Carlo simulations to estimate defect rates under uncertainty and adjusted strategies accordingly.
- Produced visualized outputs and evaluated model trade-offs between precision and computational cost.

WORK & ACTIVITY EXPERIENCE

The Chinese University of Hong Kong, Shenzhen, School of Management and Economics **Shenzhen, China**
Teaching Assistant Sep 2024 – Present
(in Microeconomics · Financial Management · Econometrics · Machine Learning)

- Held 110+ office hours in total to address 600+ student queries; graded 300+ assignments
- Led 40+ tutorials to supplement course materials with 700+ pages slides made in LaTeX; instructed data analysis in Excel; engaged in compiling and preparing the lecture materials of Econometrics throughout all semester

- Produced supplementary video lectures on financial concepts; shared via Chinese social media platforms and benefited 3,000+ viewers nationwide
- Led weekly tutorials in Microeconomics for up to 85 students per session; widely recognized by peers as the most popular and effective Undergraduate Student Teaching Fellows (USTFs) for undergraduate economics courses

CUHK-Shenzhen Economics Club, School of Management and Economics **Shenzhen, China**
Head of Academic Department; Vice President of Club Mar 2024 – Present

- Organized large-scale academic review sessions for Microeconomics and Econometrics for 3 semesters, including an English-language Microeconomics lecture attended by 400+ students, both local and international
- Designed and delivered 120+ LaTeX-compiled slides for an Econometrics session, supplementing core materials and introducing foundational STATA commands
- Coordinated 10+ paper reading seminars; Facilitated undergraduate engagement with frontier research by organizing researching seminars and presenting analytical commentary on influential works (e.g., Angrist's 1996 IV paper)

CUHK-Shenzhen English Debate Team (EDT) **Shenzhen, China**
Formal Selected Member Oct 2023 – Present

- Represented CUHK-Shenzhen in multiple intercollegiate BP-format debate tournaments, focusing on policy and economic motions, showcasing advanced understanding of economic and political theory through structured debate
- Competed in:
  - Novice Debate Challenge, 2023: Reached Semifinal; Enrolled into EDT official member
  - Macau Debate Open, 2024: Placed 1st in group on motions concerning fiscal policy, market liberalism & state capitalism
  - Greater Bay Area Novice Championship (GBANC), 2025: Awarded 4th Best Speaker (Open Group); Reached Semifinal; Placed 1st in group on motions involving microfinance, oligopolistic competition, and market regulation)
  - Guangdong Debate Challenge (GDC), 2025: Placed 1st in group on motions concerning business, companies' structure, startups/SME, central bank financing/monetary policy & tariffs

CUHK-Shenzhen SME News **Shenzhen, China**
Formal Editor; Student Reporter Sep 2023 – Present

- Reported on key SME events from an economic perspective, including Open Day forums, faculty interviews, and alumni profiles, highlighting economic insights and career trajectories
- Facilitated external communications with professors and academic departments to ensure the accurate and accessible presentation of economic research and teaching philosophy
- Interviewed professors on topics such as behavioral economics and development economics, transforming academic content into engaging stories for a student audience

The Chinese University of Hong Kong, Shenzhen, School of Management and Economics **Shenzhen, China**
Mentor, SME Academic Advisory (AA) Programme for Year 1 Students Jan 2025 – Nov 2025
Student Lecturer, 2026 Winter Camp for Career Exploration for High School Students Feb 2026

- Served in the mentor group for 100+ freshmen to help them better adapt to university lives; engaged in the foundation of the new program for students who major in economics
- Held AA meetings and shared experiences on exchange application, academic writing, and course selection
- Led one-by-one QA; collected students' advice in setting of the course and helped establish two new courses

LEADERSHIP EXPERIENCE

Office of Student Affairs, The Chinese University of Hong Kong, Shenzhen **Shenzhen, China**
The Representative of Freshmen, English Pre-sessional Course (EPC), 2023 Aug 2023
Mentor and Class Leader; The Representative of seniors, English Pre-sessional Course (EPC), 2025 Aug 2025

- Selected through competitive English assessments to serve as the official student representative in both 2023 and 2025 EPC program. Delivered keynote speeches in English to audiences of 600+ (2023) and 800+ (2025) participants across online and offline platforms; featured on official university media for outstanding communication and leadership
- As 2025 senior mentor, led a cohort of 30 freshmen through academic and campus life orientation; organized activities, shared personal learning experiences, and supported their English learning in collaboration with EPC faculty. Acted

as a liaison between students, instructors, and administrators in a bilingual academic environment, enhancing peer engagement and cross-cultural communication

Ling College, The Chinese University of Hong Kong, Shenzhen **Shenzhen, China**
The Host of the First Lesson Sep 2023
Member of the Committee of Ling College Sep 2023 – Present

- Hosted the college wide "First Lesson" event, addressing the entire residential community and welcoming new students
- Serve as an active member of the residential committee, organizing student events and promoting well-being through peer support and campus initiatives; constructed the activities like "Studying for 21 days" to encourage students develop the habit of perseverance and studing

SKILLS AND INTERESTS

- *Computer Skills:* Microsoft Office, STATA, Python (incl. NumPy, Pandas, Matplotlib, Natural Language Processing (NLP), Speech Recognition (ASR), Web Scraping, Machine Learning Method), R, LaTeX, MATLAB, Jupyter Notebook, Eviews
- *Language Skills:* Mandarin Chinese (Native), English (full professional proficiency; primary language of instruction and academic work), French (Used as the working language for the project with Prof. Clotaire Weigel*)*
- *Interests:* Writing (Awarded first prize in city-level writing competitions), Debating (with multiple Best Debater awards in both English and Mandarin competitions), Badminton, Philosophy (Feminist & Existentialist)